SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AI & DS	**Date:**	

Code of Conduct

I S.N.V.S Karthik / 192424186 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Karthik

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as "best hotels near…" and "book a flight from Chennai to…". The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as "give the book to John," "sleep," and "put the file on the desk." The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

1. Constituency Parsing vs. Dependency Parsing

Constituency parsing represents a sentence as a hierarchical tree of phrases such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). It is useful for understanding the internal structure of sentences. For example, in a sentence with a long-distance dependency, constituency parsing shows how words are grouped into larger grammatical units.

Dependency parsing, on the other hand, directly represents relationships between individual words. It identifies a **head word** and its dependents, such as subject, object, modifier, and prepositional complement. This makes relationships between words easier to extract.

Aspect	Constituency Parsing	Dependency Parsing
Main representation	Phrase hierarchy	Word-to-word relationships
Structure	Tree of phrases	Tree of dependencies
Focus	Grammatical constituents	Relationships between words
Useful for	Phrase structure analysis	Information extraction and semantic relations

For conversational AI, **dependency parsing is generally more suitable** because conversational applications need to identify who performed an action, what object was affected, where something happened, and similar relationships. It provides direct connections between words and is therefore efficient for extracting meaningful relationships from conversational text.

2. Top-Down Parsing vs. Earley Parsing

Conventional top-down parsing begins with the start symbol of the grammar and predicts possible structures until it reaches the input words. It works well with complete and relatively unambiguous sentences, but incomplete input can cause difficulties because the parser may predict structures that cannot ultimately be completed.

Earley parsing is a chart-parsing algorithm that maintains partial parsing states. It can handle **incomplete, ambiguous, and left-recursive grammars** more effectively. It does not need to completely determine the structure before processing further input.

For example:

- **"best hotels near…"** → Earley parsing can maintain a partial interpretation while waiting for the missing location.
- **"book a flight from Chennai to…"** → It can preserve the incomplete structure and continue when the destination is entered.

Aspect	Top-Down Parsing	Earley Parsing
Complete input	Good	Good
Incomplete input	Less suitable	Highly suitable
Ambiguity	Can be expensive/problematic	Handles ambiguity well
Partial structures	Limited	Explicitly maintained
Interactive applications	Less flexible	More flexible

Therefore, **Earley parsing is more appropriate for interactive NLP applications**, especially search engines, chatbots, and voice assistants where users frequently provide partial or ambiguous queries.

3. CFG vs. PCFG vs. Neural Constituency Parsing

A **Context-Free Grammar (CFG)** represents possible sentence structures using grammatical production rules. When a sentence has multiple valid parse trees, CFG can represent all of them but does not inherently determine which interpretation is more likely.

A **Probabilistic Context-Free Grammar (PCFG)** extends CFG by assigning probabilities to grammar rules. When multiple parse trees are possible, the parser can select the tree with the highest probability. Thus, PCFG provides a statistical method for resolving ambiguity while remaining relatively interpretable.

A **neural constituency parser** uses machine-learning models, often based on neural language representations, to predict syntactic structures. It can learn complex contextual patterns from large datasets and generally provides higher practical accuracy, particularly for natural and conversational language.

Approach	Ambiguity Representation	Ambiguity Resolution	Interpretability	Accuracy
CFG	Multiple parse trees	No probability-based preference	Very high	Limited
PCFG	Multiple weighted trees	Highest-probability tree	High	Better
Neural parser	Learned representations	Context-based prediction	Lower	Generally highest

For a grammar-checking system, **PCFG provides the best balance between interpretability and practical accuracy** when the goal is to clearly explain grammatical decisions.

4. Feature Structures vs. Traditional CFG

A traditional **Context-Free Grammar** mainly represents syntactic structures through production rules such as:

S → NP VP

However, ordinary CFG rules do not directly represent detailed grammatical properties such as number, gender, person, and tense.

Feature structures extend grammatical representations by associating features and values with linguistic elements. For example:

NP → [Number: Singular, Gender: Feminine]

These features can be unified with features of other constituents to enforce grammatical agreement. For example:

She walks.

The subject has:

- Person = 3rd
- Number = Singular

Therefore, the verb must also satisfy the appropriate agreement requirements.
Feature structures are particularly valuable in multilingual NLP because different languages may have different requirements for gender, number, case, person, tense, and agreement.

Aspect	**Traditional CFG**	**Feature Structures**
Basic syntax	Yes	Yes
Number	Limited	Explicit
Gender	Limited	Explicit
Person	Limited	Explicit
Agreement constraints	Difficult	Easy to enforce
Multilingual processing	Less flexible	More suitable

Therefore, **feature structures are more useful for multilingual systems** because they allow grammatical constraints to be explicitly represented and checked while retaining the structural capabilities of grammar rules.

5. Subcategorization Frames

A **subcategorization frame** specifies the types and number of arguments that a verb requires or permits. Different verbs have different argument requirements.

For example:

- **Give** → requires an agent, an object, and a recipient

  “Give the book to John.”

- **Sleep** → generally requires only a subject

  “John sleeps.”

- **Put** → requires an object and a location

  “Put the file on the desk.”

The system can therefore use verb-specific frames to determine whether a sentence contains the required arguments.

Verb	**Subcategorization Pattern**	**Example**
give	Subject + Object + Recipient	“She gave the book to John.”
sleep	Subject	“John sleeps.”
put	Subject + Object + Location	“She put the file on the desk.”

Subcategorization is important because it helps both **syntactic and semantic analysis**. Syntactically, it determines whether the sentence has the correct structure. Semantically, it helps identify the roles associated with the verb, such as agent, object, recipient, and location.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	3	3	3	3		16	75
2	3	3	3	3		16	
3	3	3	3	3		16	
4	3	3	3	3		16	
5	3	3	3	3		16	

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